

1. length()

Question

Write a Java program to find the length of a string using the `length()` predefined method.

Steps

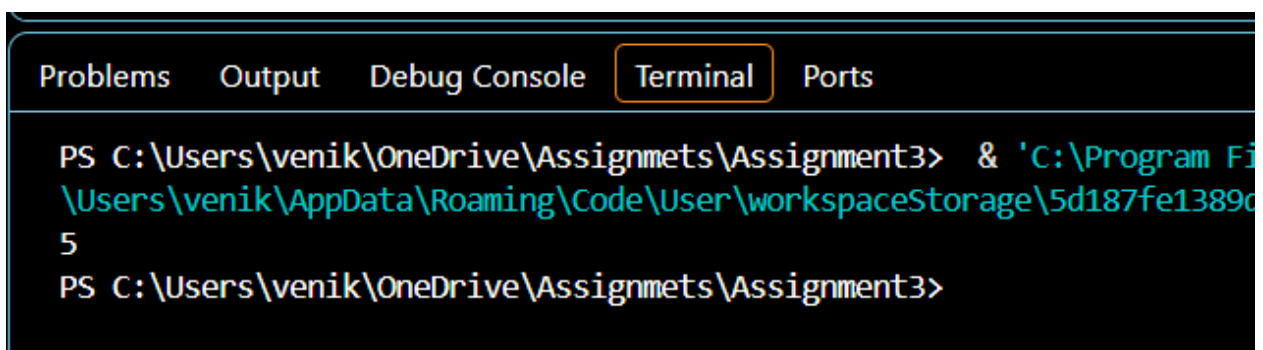
1. Start the program.
2. Create a string.
3. Use `length()` to find the number of characters.
4. Display the length.
5. Stop the program.

Algorithm

1. Start.
2. Declare `String str = "Hello".`
3. Calculate `str.length()`.
4. Print the result.
5. Stop.

Program

```
public class StringLength {  
    public static void main(String[] args) {  
        String str = "Hello";  
  
        System.out.println(str.length());  
    }  
}
```



The screenshot shows a terminal window with tabs for Problems, Output, Debug Console, Terminal, and Ports. The Terminal tab is active, displaying the command prompt and the output of a Java program. The command executed is `PS C:\Users\venik\OneDrive\Assignmets\Assignment3> java -cp 'C:\Program Files\Java\jdk-11.0.10\bin\java.exe' 'C:\Users\venik\AppData\Roaming\Code\User\workspaceStorage\5d187fe1389c\5' StringLength`. The output is `5`.

```
PS C:\Users\venik\OneDrive\Assignmets\Assignment3> java -cp 'C:\Program Files\Java\jdk-11.0.10\bin\java.exe' 'C:\Users\venik\AppData\Roaming\Code\User\workspaceStorage\5d187fe1389c\5' StringLength  
5  
PS C:\Users\venik\OneDrive\Assignmets\Assignment3>
```

2. charAt()

Question

Write a Java program to find a character at a specified position using the `charAt()` predefined method.

Steps

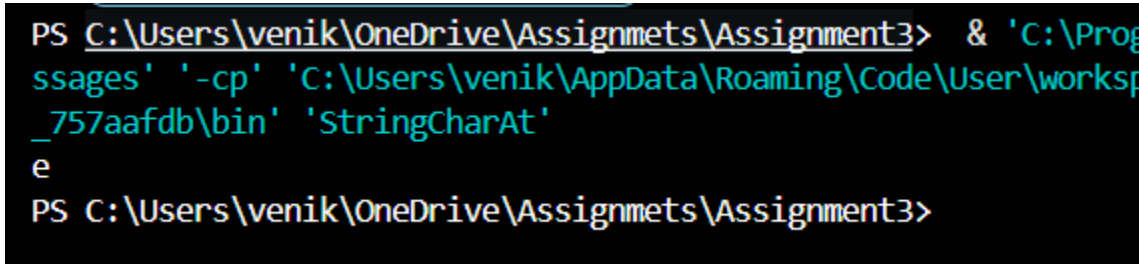
1. Start the program.
2. Create a string.
3. Specify the position of the character.
4. Use `charAt()`.
5. Display the character.
6. Stop.

Algorithm

1. Start.
2. Declare `String str = "Hello"`.
3. Use `str.charAt(1)`.
4. Print the character.
5. Stop.

Program

```
public class StringCharAt {  
    public static void main(String[] args) {  
        String str = "Hello";  
  
        System.out.println(str.charAt(1));  
    }  
}
```



```
PS C:\Users\venik\OneDrive\Assignmets\Assignment3> & 'C:\Program Files\Java\jdk-11.0.2\bin\javac.exe' -cp 'C:\Users\venik\AppData\Roaming\Code\User\workspace\757aafdb\bin' 'StringCharAt'  
PS C:\Users\venik\OneDrive\Assignmets\Assignment3>
```

3. toUpperCase()

Question

Write a Java program to convert a string into uppercase using the `toUpperCase()` predefined method.

Steps

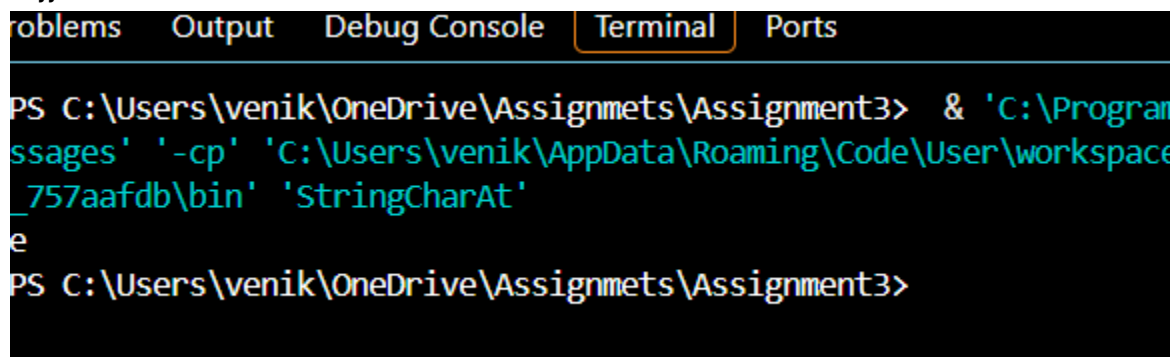
1. Start the program.
2. Create a lowercase string.
3. Use `toUpperCase()`.
4. Display the uppercase string.
5. Stop.

Algorithm

1. Start.
2. Declare `String str = "hello"`.
3. Apply `str.toUpperCase()`.
4. Print the result.
5. Stop.

Program

```
public class StringUpperCase {  
    public static void main(String[] args) {  
        String str = "hello";  
  
        System.out.println(str.toUpperCase());  
    }  
}
```



```
Problems  Output  Debug Console  Terminal  Ports  
PS C:\Users\venik\OneDrive\Assignmets\Assignment3> & 'C:\Program  
ssages' '-cp' 'C:\Users\venik\AppData\Roaming\Code\User\workspace  
_757aafdb\bin' 'StringCharAt'  
e  
PS C:\Users\venik\OneDrive\Assignmets\Assignment3>
```

4. toLowerCase()

Question

Write a Java program to convert a string into lowercase using the `toLowerCase()` predefined method.

Steps

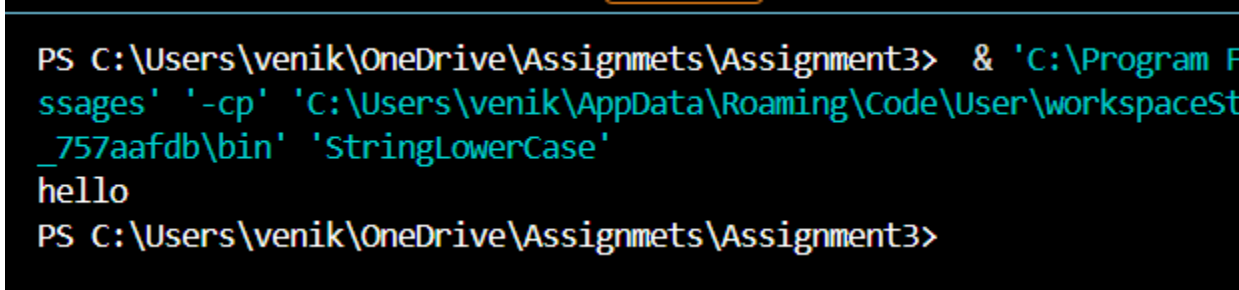
1. Start the program.
2. Create an uppercase string.
3. Use `toLowerCase()`.
4. Display the lowercase string.
5. Stop.

Algorithm

1. Start.
2. Declare `String str = "HELLO"`.
3. Apply `str.toLowerCase()`.
4. Print the result.
5. Stop.

Program

```
public class StringLowerCase {  
    public static void main(String[] args) {  
        String str = "HELLO";  
  
        System.out.println(str.toLowerCase());  
    }  
}
```



```
PS C:\Users\venik\OneDrive\Assignmets\Assignment3> & 'C:\Program F  
ssages' '-cp' 'C:\Users\venik\AppData\Roaming\Code\User\workspaceSt  
_757aafdb\bin' 'StringLowerCase'  
hello  
PS C:\Users\venik\OneDrive\Assignmets\Assignment3>
```

5. equals()

Question

Write a Java program to compare two strings using the `equals()` predefined method.

Steps

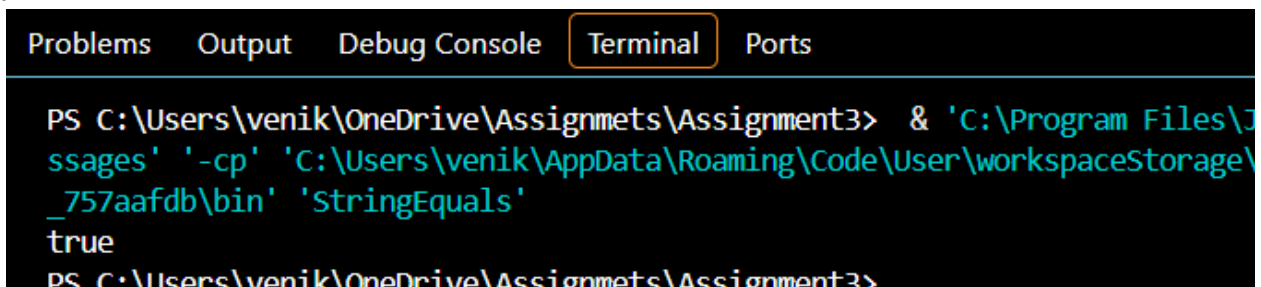
1. Start the program.
2. Create two strings.
3. Compare them using `equals()`.
4. Display the result.
5. Stop.

Algorithm

1. Start.
2. Declare two strings.
3. Use `str1.equals(str2)`.
4. Print the result.
5. Stop.

Program

```
public class StringEquals {  
    public static void main(String[] args) {  
        String str1 = "Hello";  
        String str2 = "Hello";  
  
        System.out.println(str1.equals(str2));  
    }  
}
```



The screenshot shows a terminal window with tabs for Problems, Output, Debug Console, Terminal, and Ports. The Terminal tab is active, displaying the command to run the program and its output. The command is: `PS C:\Users\venik\OneDrive\Assignmets\Assignment3> & 'C:\Program Files\Java\jdk-11.0.2\bin\java.exe' -cp 'C:\Users\venik\AppData\Roaming\Code\User\workspaceStorage\757aafdb\bin' 'StringEquals'`. The output is `true`.

```
PS C:\Users\venik\OneDrive\Assignmets\Assignment3> & 'C:\Program Files\Java\jdk-11.0.2\bin\java.exe' -cp 'C:\Users\venik\AppData\Roaming\Code\User\workspaceStorage\757aafdb\bin' 'StringEquals'  
true  
PS C:\Users\venik\OneDrive\Assignmets\Assignment3>
```

6. equalsIgnoreCase()

Question

Write a Java program to compare two strings without considering their case using `equalsIgnoreCase()`.

Steps

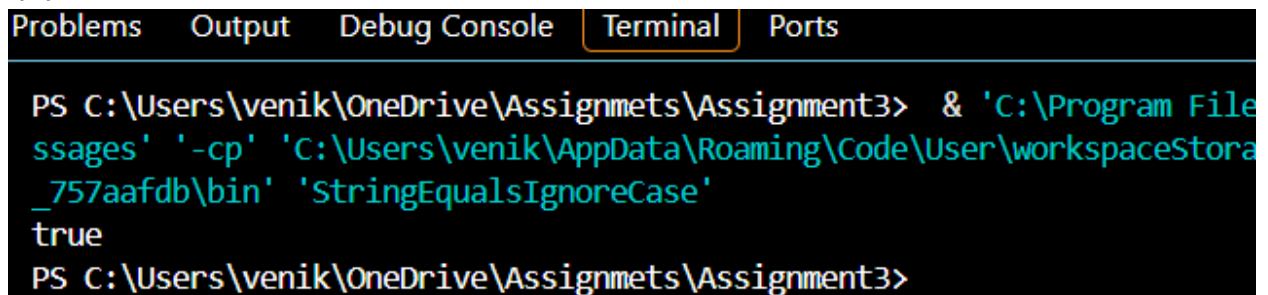
1. Start the program.
2. Create two strings with different cases.
3. Use `equalsIgnoreCase()`.
4. Display the result.
5. Stop.

Algorithm

1. Start.
2. Declare two strings.
3. Apply `str1.equalsIgnoreCase(str2)`.
4. Print the result.
5. Stop.

Program

```
public class StringEqualsIgnoreCase {  
    public static void main(String[] args) {  
        String str1 = "Hello";  
        String str2 = "HELLO";  
  
        System.out.println(str1.equalsIgnoreCase(str2));  
    }  
}
```



The screenshot shows a terminal window with tabs for Problems, Output, Debug Console, Terminal, and Ports. The Terminal tab is active, displaying the command to run the program and its output. The command is `PS C:\Users\venik\OneDrive\Assignmets\Assignment3> & 'C:\Program Files\Microsoft Visual Studio\2019\Community\MSBuild\Tools\MSBuild.exe' 'C:\Users\venik\AppData\Roaming\Code\User\workspaceStorage\757aafdb\bin' 'StringEqualsIgnoreCase'`, and the output is `true`.

```
PS C:\Users\venik\OneDrive\Assignmets\Assignment3> & 'C:\Program Files  
ssages' '-cp' 'C:\Users\venik\AppData\Roaming\Code\User\workspaceStora  
_757aafdb\bin' 'StringEqualsIgnoreCase'  
true  
PS C:\Users\venik\OneDrive\Assignmets\Assignment3>
```

7. concat()

Question

Write a Java program to join two strings using the `concat()` predefined method.

Steps

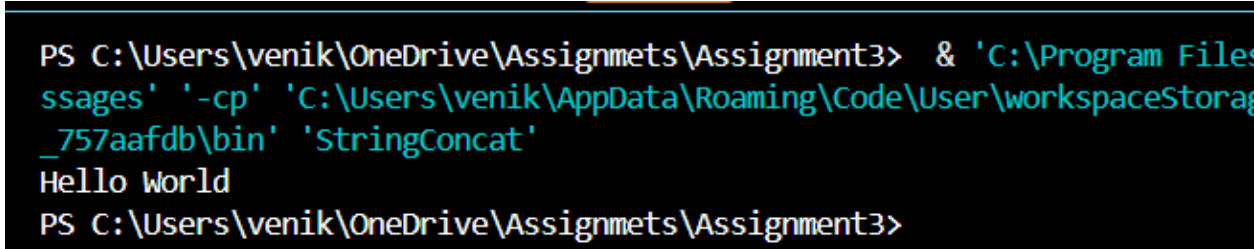
1. Start the program.
2. Create two strings.
3. Use `concat()` to join them.
4. Display the combined string.
5. Stop.

Algorithm

1. Start.
2. Declare two strings.
3. Use `str1.concat(str2)`.
4. Print the result.
5. Stop.

Program

```
public class StringConcat {  
    public static void main(String[] args) {  
        String str1 = "Hello ";  
        String str2 = "World";  
  
        System.out.println(str1.concat(str2));  
    }  
}
```



```
PS C:\Users\venik\OneDrive\Assignmets\Assignment3> & 'C:\Program Files  
ssages' '-cp' 'C:\Users\venik\AppData\Roaming\Code\User\workspaceStorage  
_757aafdb\bin' 'StringConcat'  
Hello World  
PS C:\Users\venik\OneDrive\Assignmets\Assignment3>
```

8. substring()

Question

Write a Java program to extract a part of a string using the `substring()` predefined method.

Steps

1. Start the program.
2. Create a string.
3. Specify the starting position.
4. Use `substring()`.
5. Display the extracted string.
6. Stop.

Algorithm

1. Start.
2. Declare `String str = "Hello World"`.
3. Apply `str.substring(6)`.
4. Print the result.
5. Stop.

Program

```
public class StringSubstring {  
    public static void main(String[] args) {  
        String str = "Hello World";  
  
        System.out.println(str.substring(6));  
    }  
}
```

```
PS C:\Users\venik\OneDrive\Assignmets\Assignment3> & 'C:\P  
ssages' '-cp' 'C:\Users\venik\AppData\Roaming\Code\User\wor  
_757aafdb\bin' 'StringSubstring'  
World  
PS C:\Users\venik\OneDrive\Assignmets\Assignment3>
```

}

9. replace()

Question

Write a Java program to replace a character in a string using the `replace()` predefined method.

Steps

1. Start the program.
2. Create a string.
3. Specify the character to replace.
4. Specify the new character.
5. Display the modified string.
6. Stop.

Algorithm

1. Start.
2. Declare `String str = "Hello".`
3. Use `str.replace('l', 'x').`
4. Print the result.
5. Stop.

Program

```
public class StringReplace {  
    public static void main(String[] args) {  
        String str = "Hello";  
  
        System.out.println(str.replace('l', 'x'));  
    }  
}
```

```
PS C:\Users\venik\OneDrive\Assignmets\Assignment3> & 'ssages' '-cp' 'C:\Users\venik\AppData\Roaming\Code\User_757aafdb\bin' 'StringReplace'  
Hexxo  
PS C:\Users\venik\OneDrive\Assignmets\Assignment3>
```

```
}
```

10. contains()

Question

Write a Java program to check whether a string contains a particular sequence using the `contains()` predefined method.

Steps

1. Start the program.
2. Create a string.
3. Specify the text to search.
4. Use `contains()`.
5. Display the result.
6. Stop.

Algorithm

1. Start.
2. Declare `String str = "Hello World"`.
3. Use `str.contains("World")`.
4. Print the result.
5. Stop.

Program

```
public class StringContains {  
    public static void main(String[] args) {  
        String str = "Hello World";  
  
        System.out.println(str.contains("World"));  
    }  
}
```

```
PS C:\Users\venik\OneDrive\Assignmets\Assignment3> & 'C:\Program  
Files\Java\jdk-11.0.2\bin\java.exe' -cp 'C:\Users\venik\AppData\Roaming\Code\User\workspace  
_757aafdb\bin' 'StringContains'  
true  
PS C:\Users\venik\OneDrive\Assignmets\Assignment3>
```

```
}
```

11. indexOf()

Question

Write a Java program to find the first occurrence of a character using the `indexOf()` predefined method.

Steps

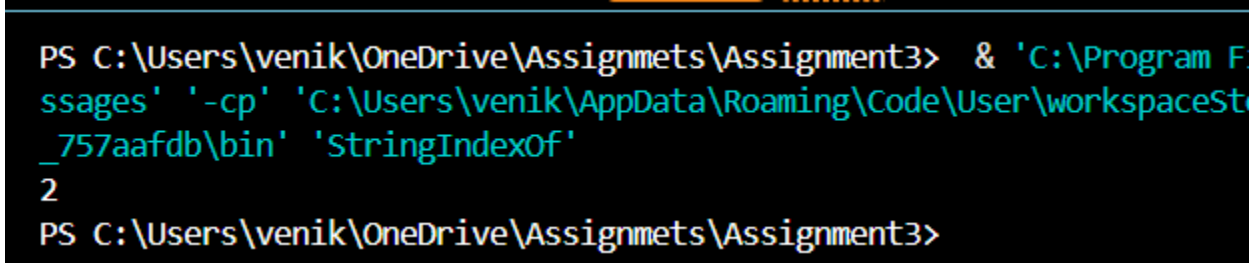
1. Start the program.
2. Create a string.
3. Specify the character to search.
4. Use `indexOf()`.
5. Display its position.
6. Stop.

Algorithm

1. Start.
2. Declare `String str = "Hello"`.
3. Use `str.indexOf('l')`.
4. Print the position.
5. Stop.

Program

```
public class StringIndexOf {  
    public static void main(String[] args) {  
        String str = "Hello";  
  
        System.out.println(str.indexOf('l'));  
    }  
}
```



```
PS C:\Users\venik\OneDrive\Assignmets\Assignment3> & 'C:\Program Files\Java\jdk-11.0.2\bin\javac.exe' -cp 'C:\Users\venik\AppData\Roaming\Code\User\workspaceStorage\757aafdb\bin' 'StringIndexOf'  
2  
PS C:\Users\venik\OneDrive\Assignmets\Assignment3>
```

12. `lastIndexOf()`

Question

Write a Java program to find the last occurrence of a character using the `lastIndexOf()` predefined method.

Steps

1. Start the program.
2. Create a string.
3. Specify the character to search.
4. Use `lastIndexOf()`.
5. Display its position.
6. Stop.

Algorithm

1. Start.
2. Declare `String str = "Hello"`.
3. Use `str.lastIndexOf('l')`.
4. Print the position.
5. Stop.

Program

```
public class StringLastIndexOf {  
    public static void main(String[] args) {  
        String str = "Hello";  
  
        System.out.println(str.lastIndexOf('l'));  
    }  
}
```

```
PS C:\Users\venik\OneDrive\Assignmets\Assignment3> & 'C:\Program Files\Java\jdk-25.0.3\bin\java.exe'  
ssages' '-cp' 'C:\Users\venik\AppData\Roaming\Code\User\workspaceStorage\5d187fe1389deb9b54384595ef8a  
_757aafdb\bin' 'StringLastIndexOf'  
3  
PS C:\Users\venik\OneDrive\Assignmets\Assignment3>
```

13. `startsWith()`

Question

Write a Java program to check whether a string starts with a particular word using `startsWith()`.

Steps

1. Start the program.
2. Create a string.
3. Specify the starting text.
4. Use `startsWith()`.
5. Display the result.
6. Stop.

Algorithm

1. Start.
2. Declare `String str = "Hello World"`.
3. Use `str.startsWith("Hello")`.
4. Print the result.
5. Stop.

Program

```
public class StringStartsWith {  
    public static void main(String[] args) {  
        String str = "Hello World";  
  
        System.out.println(str.startsWith("Hello"));  
    }  
}
```

```
PS C:\Users\venik\OneDrive\Assignmets\Assignment3> & 'C:\Program  
ssages' '-cp' 'C:\Users\venik\AppData\Roaming\Code\User\workspaces  
_757aafdb\bin' 'StringStartsWith'  
true  
PS C:\Users\venik\OneDrive\Assignmets\Assignment3>  
}
```

14. endsWith()

Question

Write a Java program to check whether a string ends with a particular word using `endsWith()`.

Steps

1. Start the program.
2. Create a string.
3. Specify the ending text.
4. Use `endsWith()`.
5. Display the result.
6. Stop.

Algorithm

1. Start.
2. Declare `String str = "Hello World"`.
3. Use `str.endsWith("World")`.
4. Print the result.
5. Stop.

Program

```
public class StringEndsWith {  
    public static void main(String[] args) {  
        String str = "Hello World";  
  
        System.out.println(str.endsWith("World"));  
    }  
}
```

```
PS C:\Users\venik\OneDrive\Assignmets\Assignment3> & 'C:\Program Files\Java\jdk-25.0.3\bin\java.exe' %ssages' -cp 'C:\Users\venik\AppData\Roaming\Code\User\workspaceStorage\5d187fe1389deb9b54384595e_757aafdb\bin' 'StringEndsWith'  
true  
PS C:\Users\venik\OneDrive\Assignmets\Assignment3>
```

15. trim()

Question

Write a Java program to remove the spaces at the beginning and end of a string using the `trim()` predefined method.

Steps

Start the program.

Create a string with spaces.

Use `trim()`.

Display the string after removing spaces.

Stop.

Algorithm

1. Start.
2. Declare `String str = " Hello "`.
3. Apply `str.trim()`.
4. Print the result.
5. Stop.

Program

```
public class StringTrim {  
    public static void main(String[] args) {  
        String str = " Hello ";  
  
        System.out.println(str.trim());  
    }  
}
```

```
PS C:\Users\venik\OneDrive\Assignmets\Assignment3> & 'C:\Program Files\Java\jdk-25.0.3\bin\java.exe' '-X  
ssages' '-cp' 'C:\Users\venik\AppData\Roaming\Code\User\workspaceStorage\5d187fe1389deb9b54384595ef8ad6ff  
_757aafdb\bin' 'StringTrim'  
Hello  
PS C:\Users\venik\OneDrive\Assignmets\Assignment3>
```